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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT347

Course Name: ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What is civic virtue? How does it relate to professional ethics?	1	(3)
2		What do you mean by 'gendered technology' and provide one specific example of a product whose design or application reflects this concept.	2	(3)
3		Explain the term triple bottom line in sustainable engineering.	3	(3)
4		Why is biodiversity considered essential for ecosystem stability? Give one example.	3	(3)
5		What is the idea behind degrowth? How is it different from economic development?	4	(3)
6		Identify two engineering solutions that can reduce greenhouse gas emissions in everyday life.	5	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9		You are part of a team designing an app that collects user data for public services. What ethical concerns would you raise regarding data privacy and consent? Suggest safeguards.	1	(8)
10	a)	What is the fundamental difference between 'Equity' and 'Equality' when discussing gender initiatives and resource allocation in education/workplace?	2	(5)
	b)	Suggest measures to promote gender equity in workplaces.	2	(3)

Module 2

11		A new industrial project near a wetland promises jobs but threatens local ecosystems. Apply the three ethical lenses (anthropocentric, biocentric, ecocentric) to assess it and recommend a balanced decision.	3	(8)
12	a)	Explain how life-cycle analysis (LCA) can help engineers design more sustainable products.	3	(5)
	b)	Illustrate with one example relevant to Kerala.	3	(3)

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Module 3				
13		Rapid motorization is worsening air pollution. Suggest sustainable mobility options suitable for medium-sized Kerala towns, explaining their environmental benefits.	4	(8)
14		Outline a three-step plan to move a college campus towards zero waste. Mention the role of engineers, students, and the college management in each step.	4	(8)
Module 4				
15	a)	Discuss how renewable energy technologies (solar, wind, biogas) can be scaled up in Kerala.	5	(4)
	b)	State two technical and two social challenges to their adoption.	5	(4)
16	a)	Discuss the importance of environmental policies and regulations in promoting sustainable engineering.	5	(4)
	b)	Give one national and one international example and their impact.	5	(4)
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ANSWER KEY – ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

PART A (3 Marks Each)

1. What is civic virtue? How does it relate to professional ethics?

Civic virtue refers to behaviours and attitudes that support the functioning of a democratic society—such as participating in civic duties, obeying laws, voting, volunteering, and prioritizing common good over personal gain (Module 1).

Relation to Professional Ethics:
Professionals, especially engineers, are expected to act with responsibility toward society. Civic virtue reinforces professionalism by encouraging ethical conduct, accountability, fairness, respect for law, and service to the public. A professional practising civic virtue ensures their work enhances social welfare and public safety.

2. What do you mean by 'gendered technology'? Provide an example.

Gendered technology refers to technologies whose design, usage, assumptions, or impacts reflect gender stereotypes or biases (Module 1). Technology can unintentionally reinforce unequal gender roles.

Example:
Virtual assistants like Siri/Alexa often use female voices, reinforcing stereotypes that women are suited for "assistant" or "helping" roles. Another example is facial recognition systems that perform poorly for female faces due to biased datasets.

3. Explain the term triple bottom line in sustainable engineering.

The **Triple Bottom Line (TBL)** is a sustainability framework that evaluates engineering decisions on **three dimensions** (Module 2):

1. **Economic sustainability** – affordability, cost-efficiency, long-term savings.
2. **Social sustainability** – human welfare, equity, safety, inclusion.
3. **Environmental sustainability** – conserving ecosystems, reducing pollution, minimizing carbon footprint.

Sustainable engineering requires designing solutions that balance **people, planet, and profit**.

4. Why is biodiversity essential for ecosystem stability? Give an example.

Biodiversity ensures that ecosystems remain stable, resilient, and capable of withstanding disturbances (Module 2).
High biodiversity provides:

- Balanced food webs

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- Disease resistance
- Ecological services such as pollination, oxygen production, soil fertility

Example:
Western Ghats in Kerala—home to endemic species like Nilgiri Tahr and Lion-Tailed Macaque—maintain climatic stability, water regulation, and rainforest balance.

5. What is the idea behind degrowth? How is it different from economic development?

Degrowth advocates reducing excessive consumption and production in wealthy societies to achieve ecological balance, social justice, and sustainability (Module 3). It promotes "sufficiency over efficiency."

Difference:

- **Economic development** focuses on continuous growth of GDP and industrial output.
- **Degrowth** challenges infinite growth on a finite planet and emphasizes well-being, equality, and environmental protection over material expansion.

6. Identify two engineering solutions that reduce greenhouse gas emissions in everyday life.

1. **Renewable energy solutions** – rooftop solar, solar water heaters, biogas for cooking.
2. **Sustainable mobility** – electric vehicles, public transport like Kochi Water Metro, cycling paths, and e-buses.

Other valid points: energy-efficient appliances, green buildings, smart grids.

PART B (8 Marks Each)

Module 1

9. Ethical concerns in an app collecting user data + safeguards.

Ethical Concerns (Module 1):

1. **Privacy violation** – collecting more data than necessary.
2. **Lack of informed consent** – users may not understand how data will be used.
3. **Data security risks** – unauthorized access, hacking.
4. **Misuse of personal information** – sharing with third parties without consent.
5. **Surveillance concerns** – potential monitoring of behaviour.
6. **Bias and discrimination** – algorithmic misuse of data.

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Safeguards:

- **Informed consent** with clear explanations of data use.
- **Data minimization** – collect only what is necessary.
- **Strong encryption and cybersecurity measures.**
- **Transparent data policy** aligned with cybertrust principles.
- **Options for user control** – ability to delete or modify data.
- **Regular audits** and compliance with laws like IT Act/GDPR standards.

10(a). Equity vs Equality in gender initiatives (5 marks)

Equality:
Providing the same resources or opportunities to everyone (Module 1).
Example: Giving the same scholarship amount to all students.

Equity:
Allocating resources based on individual needs to ensure fair outcomes.
Example: Providing extra support for girls in STEM fields where they are underrepresented.

Difference:
Equality assumes all start from same position; equity recognises systemic biases and seeks fairness.

10(b). Measures to promote gender equity in workplaces (3 marks)

- Gender-sensitive recruitment and promotion policies.
- Flexible work hours and maternity/paternity leave.
- Anti-harassment committees and grievance redressal (POSH Act).
- Safe, inclusive workspaces and gender-sensitivity training.
- Address algorithmic bias in technological systems.

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Module 2

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11. Using Anthropocentric, Biocentric, Ecocentric lenses on an industrial project near a wetland.

Anthropocentric View:
Humans are central. Benefits like jobs, economic growth, and infrastructure may justify the project. However, long-term human welfare (fisheries, flood control) depends on a healthy wetland.

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Biocentric View:
All living organisms have intrinsic value. The project threatens species, migratory birds, fish nurseries, and biodiversity. The project should be halted or redesigned to protect life forms.

Ecocentric View:
The entire ecosystem—including soil, water, flora, fauna—has value. Wetlands regulate floods, groundwater recharge, climate moderation. Disturbing them destabilizes the entire system.

Balanced Recommendation:

- Allow project only with strict Environmental Impact Assessment (EIA).
- Maintain environmental flow, buffer zones, and waste-treatment systems.
- Consider shifting location or adopting eco-friendly design.
- Ensure community participation and compensation.

12(a). How LCA supports sustainable product design (5 marks)

Life Cycle Analysis

- Evaluates environmental impact from **cradle to grave** (raw material → manufacturing → transport → use → disposal).
- Helps identify high-emission stages.
- Guides material substitution, energy savings, eco-design.
- Supports certifications, eco-labelling, and compliance with sustainability regulations.
- Avoids shifting pollution from one stage to another.

12(b). Kerala-related example (3 marks)

Example: Coconut-based products.
LCA shows coconut shells and husk (coir pith) can be reused in:

- soil conditioners
- biofuel
- coir products

This supports Kerala's **circular economy** and reduces landfill waste.

Or example: life-cycle analysis of plastic vs. cloth bags in Kerala supermarkets.

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Module 3

13. Sustainable mobility options for medium-sized Kerala towns & benefits (8 marks)

1. Improved Public Transport

- E-buses, mini-buses, Water Metro–style inland transport.
- **Benefits:** Reduces congestion, emissions, and fuel consumption.

2. Non-Motorized Transport (NMT)

- Dedicated cycling lanes, walkable streets.
- **Benefits:** Zero emissions, better health, low infrastructure cost.

3. Electric Mobility

- E-autos and EV charging stations.
- **Benefits:** Zero tailpipe emissions; reduces PM2.5 and NOx.

4. Feeder Services + Transit-Oriented Development (TOD)

- Local shuttle services feeding metro/bus hubs.
- **Benefits:** Reduces private vehicle dependence.

5. Intelligent Transport Systems (ITS)

- Smart traffic signals, real-time transit info.

6. Water-based Transport (Kochi Water Metro model)

- Suitable for towns near rivers/lakes.

Environmental Benefits:

- Lower carbon footprint
- Improved air quality
- Reduced noise pollution
- Lower fossil fuel dependency
- Helps climate change mitigation

14. Three-step plan to move a college campus toward zero waste

Step 1: Waste Segregation & Awareness

- *Engineers:* Design bin systems, waste-tracking apps, MRF layout.
- *Students:* Perform segregation, organize awareness drives, green clubs.

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- *Management:* Install bins, enforce policies, provide funds.

Step 2: Waste Reduction & Reuse Practices

- *Engineers:* Promote reusable materials, develop composting units.
- *Students:* Reuse notebooks, avoid single-use plastics.
- *Management:* Implement digital notices, bulk purchasing to reduce packaging.

Step 3: Recycling & Composting Infrastructure

- *Engineers:* Build compost pits, biogas units, E-waste collection systems.
- *Students:* Maintain compost units, volunteer for recycling campaigns.
- *Management:* Set up tie-ups with recycling agencies, adopt green protocols.

Module 4

15(a). Scaling up solar, wind, biogas in Kerala (4 marks)

Solar:

- Expand rooftop solar through Soura Scheme.
- Promote BIPV in buildings.
- Solarizing public transport (Water Metro model).

Wind:

- Utilize coastal/windy regions (Kanyakumari belt extension, Palakkad gap).
- Install hybrid solar-wind farms.

Biogas:

- Expand household and institutional biogas plants.
- Use food waste from markets & campuses.
- Support bio-CNG for transport.

15(b). Two technical & two social challenges (4 marks)

Technical Challenges:

- Intermittent supply of solar/wind.
- High initial setup cost.
- Limited land availability for solar farms.

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- Battery storage technology constraints.

Social Challenges:

- Resistance to land acquisition.
- Lack of awareness about renewable benefits.
- Visual/noise concerns (wind turbines).
- Behavioural resistance in adopting new technologies.

16(a). Importance of environmental policies & regulations (4 marks)

Policies ensure:

- Pollution control (EPA 1986).
- Sustainable use of natural resources.
- Safe industrial practices (EIA norms).
- Climate mitigation strategies (NAPCC).
- Protection of vulnerable ecosystems (CRZ rules).
- Accountability of polluting industries (ISO 14001).

Policies guide engineers to design responsibly, avoid harm, and follow preventive ethics.

16(b). One national + one international example with impact (4 marks)

National – Plastic Waste Management Rules (2016, 2021):

- Bans single-use plastics.
- Enforces Extended Producer Responsibility (EPR).
- Impact: Reduced plastic litter, promotes recycling industry.

International – Paris Agreement (2015):

- Global commitment to limit warming to 1.5–2°C.
- India pledged to reduce emission intensity by 45% by 2030.
- Impact: Boosted renewable energy adoption, climate adaptation policies worldwide.

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